AYMANE AIT DADS

AI Vision Engineer | Industrial Image Analysis · Computer Vision · Algorithm Development

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PROFESSIONAL SUMMARY

Ranked 1st in the EURECOM class leaderboard on an international AI-generated image detection benchmark (0.791 AUC, 250K training images, submitted to NTIRE 2026 @ CVPR), demonstrating end-to-end ownership of image acquisition, model design, and evaluation pipeline. At Orange Maroc, built and deployed a Random Forest + K-Means classification system on 100K+ network records, reducing manual analysis time by 60% - translating complex algorithmic output into actionable maintenance recommendations for non-technical stakeholders. Brings combined expertise in computer vision (CLIP ViT, CNN, DenseNet), image classification, feature engineering, and algorithm evaluation directly aligned with Viscon Group's industrial vision and agri-tech automation requirements. Aims to apply practical image-acquisition know-how and neural network benchmarking experience to design robust, repeatable vision solutions for fruit ripeness detection, egg inspection, and plant health monitoring in automated production lines.

CORE COMPETENCIES

Industrial Vision Systems | Image Acquisition & Analysis | Algorithm Development | Neural Network Selection | Image Classification | Feature Engineering | System Requirements Definition | Algorithm Benchmarking

WORK EXPERIENCE

Orange Maroc
Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Built and evaluated **Random Forest + K-Means classification pipeline** on 100K+ fiber-optic network records, mapping customer performance profiles to 5 commercial service tiers used directly by the network optimization team.
- Reduced manual analysis time by **~60%** through automated feature engineering, model evaluation workflows, and reproducible pipeline design on structured production data.
- Delivered stakeholder presentations translating **algorithmic model outputs** into concrete maintenance recommendations, bridging technical findings and operational decisions.
- Developed **unsupervised clustering** methodology to segment thousands of network nodes by upload speed, latency, and jitter - enabling systematic quality classification across the full node inventory.

PROJECTS

AI-Generated Image Detection - EURECOM ImSecu + NTIRE 2026 @ CVPR
EURECOM Course + International Benchmark · 2025 · Ranked 1st in Class · 0.791 AUC

- Ranked **1st in EURECOM class** on a private Kaggle leaderboard (0.791 AUC) by fine-tuning a CLIP ViT-L/14 backbone (428M params, 400M image-text pairs) with a custom MLP classification head trained on 250K images spanning 25+ generator types; also submitted to **NTIRE 2026 @ CVPR CodaBench** international benchmark.
- Benchmarked unfreezing strategies (4/6/8 transformer blocks), dual learning rates (2e-6 backbone, 5e-4 head), and ensemble methods (3-model AUC-weighted, 10-view TTA) - discovering that **1-epoch training (0.793 AUC) outperformed 4-epoch (0.767 AUC)** due to superior out-of-distribution generalization.

PyTorch · OpenCLIP · ViT · FP16 Mixed Precision · Kaggle GPU

Aerial Cactus Detection - Endangered Species Classification
EURECOM · 2025 · 99.8% Accuracy · F1 = 0.999

- Classified **17,500+ drone images** of endangered cactus species, achieving 99.8% accuracy, F1 = 0.999, and ROC AUC = 1.0 by systematically benchmarking CNN, DenseNet121, and hybrid CNN-Transformer architectures on aerial imagery.
- Conducted **controlled algorithm benchmarking** across three model families on overhead image data, directly paralleling industrial vision workflows where traditional and neural-network approaches must be evaluated before deployment.

PyTorch · DenseNet121 · CNN · Transformer · Scikit-learn

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM
Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, Computer Vision, Image Security, Statistics, NLP, Cloud Computing, Web Applications

CPGE - Mathematics & Physics - Ibn Timiya
2021 - 2023

SKILLS

Computer Vision & Image Analysis: PyTorch | OpenCLIP | CLIP ViT | CNN | DenseNet121 | Image Classification | TTA Ensembling

ML Frameworks & Algorithm Development: Scikit-learn | TensorFlow | Hugging Face Transformers | Feature Engineering | Clustering | Ensemble Methods

Data & Tooling: Python | Pandas | NumPy | Power BI | Git | Linux | Kaggle GPU | FP16 Mixed Precision